

Vonage Demo-World: how it is built

Demo-World is the field team's live demo playground: one site where a Vonage employee signs in, picks a customer scenario and runs a real Vonage API against their own phone in seconds. Every demo except the Network APIs and Identity simulators calls the production Vonage platform. This document describes the architecture, the languages and services, the design system, the webhook plumbing, the deployment pipeline and the security controls of the platform.

LIVE URL

<https://d7gj9x4l1nnrl.cloudfront.net/demoworld>

RUNTIME

Node.js 22 + Express on
AWS Elastic Beanstalk
(eu-north-1), behind
Amazon CloudFront

STATE

Amazon S3 JSON
documents, in-memory
cache; no database

CATALOGUE

46 tiles: 37 customer demos, 2 integrations, 7
internal tools, across 9 categories

**VONAGE PRODUCTS
USED LIVE**

Messages (RCS,
WhatsApp, SMS, Viber),
Verify v2, Voice, Video,
Number Insight, Client
SDK, WhatsApp Calling

AI

Amazon Bedrock
(Claude) for copy and
flows, Google Gemini 2.5
Flash for vision and
image generation

01 System architecture

CLIENT

Demo-World SPA

One HTML page, vanilla JavaScript, data-driven DEMOS catalogue. Login page, admin page, Template Creator, video room, webviews for RCS demos.

Employee's phone

Receives the demo over RCS, WhatsApp, SMS, Viber or a voice call. Replies and button taps flow back through Vonage webhooks.

Customer's screen

The same page is projected in the meeting; QR codes open the conversational demos on any phone.

EDGE AND APPLICATION

Amazon CloudFront

TLS termination, HTTPS only, caching of static assets. The `/dw/*` and `/demoworld` paths are uncached and pass every HTTP method so API calls and webhooks are always live.

Express application (server.js)

Single Node.js process on Elastic Beanstalk behind nginx. Hosts the Demo-World routes (`/dw/api/*`), the auth routes, the webhook receivers (`/2s/webhooks/*`), voice NCCO endpoints and the self-contained demo modules (Peak Season, Wayfair, ePARK, Avon).

Amazon S3

State: users, agents, message logs, conversations, per-demo data as JSON under one prefix. Assets: images, videos, PDFs and shop pages served public-read from a second prefix.

PLATFORMS

Vonage APIs

Messages API (RCS agent, WhatsApp number, SMS, Viber service), Verify v2, Voice API (NCCO, TTS, SIP), Video API (application auth, OpenTok SDK), Number Insight, Client SDK for in-browser calls, WhatsApp Business Calling.

AI services

Amazon Bedrock (Claude) drafts template copy, demo flows and call scripts. Gemini 2.5 Flash reads photos and generates card imagery for the branded demos.

Vonage Cloud Runtime satellites

Independent VCR instances the platform calls or forwards to: Avon skin analysis, WhatsApp Opt-Out Manager, Sales Hub and HubSpot bridges, video hosts.

The application is stateless apart from short-lived in-memory caches (video session ids, S3 document cache, webhook de-duplication window), so a redeploy or instance replacement loses nothing. Everything durable lives in S3.

02 Languages, frameworks and tooling

LAYER	TECHNOLOGY	NOTES
Backend	JavaScript on Node.js 22, Express 4	Single entry point <code>server.js</code> plus per-demo modules (<code>peak.js</code> , <code>wayfair.js</code> , others). Libraries: jsonwebtoken (RS256 app tokens, HS256 session tokens), AWS SDK v3 (S3, Bedrock, CloudFront, Elastic Beanstalk), ws, multer, cors, node-fetch.
Front end	HTML5, CSS3, vanilla ES2020 JavaScript	No framework, no build step. The catalogue is a JavaScript array; adding a demo is adding one object. Webviews and landing pages are standalone HTML files in <code>public/</code> .
Voice	NCCO (JSON call control)	Answer webhooks return NCCO arrays: talk, connect (PSTN, SIP, conversation), record, input.
Video	OpenTok JS SDK, Vonage Video REST	Application-based authentication; client tokens are app JWTs with session scope and ACL claims.
Tooling	Python 3 (boto3, Pillow, imageio-ffmpeg), headless Chrome	Deploy scripts, image and video rendering for demo assets, PDF and mockup generation, syntax gate before every deploy.
Infrastructure	AWS Elastic Beanstalk, CloudFront, S3, Bedrock; Vonage Cloud Runtime	One environment for the whole platform; VCR instances for services that need their own Vonage application or public URL.

03 Design system and front-end structure

Brand tokens

Real Vonage brand assets: violet `#871EFF` , lavender `#CBA1FA` , black `#0D0D0D` , playful accents coral, pink and green from developer.vonage.com. Display type Space Grotesk, body type Inter, Vonage wordmark in white and black variants served from S3. Demo webviews built in 2026 follow a black and white vector treatment with the Vonage symbol.

Page structure

- **Catalogue grid** filtered by category chips: Messaging, Verify and 2FA, Voice, Video, Number Insight, Network APIs, Identity, plus Integrations and Internal tools reached through their own links.
- **Demo page:** customer and use case, numbered instructions, a Run panel whose mode is declared per tile: `receive` (send to my phone), `converse` (QR to WhatsApp or RBM deep link), `verify` , `call` , `webcall` , `wacall` , `video` , `aicall` , `bridge` , `riskflow` , `insight` , `mock` , `tool` , `install` .
- **Template Creator:** live phone preview, WhatsApp and RCS toggle, AI draft, submit to Meta approval or send an RCS test.
- **Mock simulators** for Network APIs and Identity show endpoint, request and response JSON with a verdict pill, so the field team can demo capabilities that need carrier onboarding.
- **Per-user rendering:** the page is served with `Cache-Control: no-store` and checks the session before rendering; the admin page is a separate route.

Operational versus mock

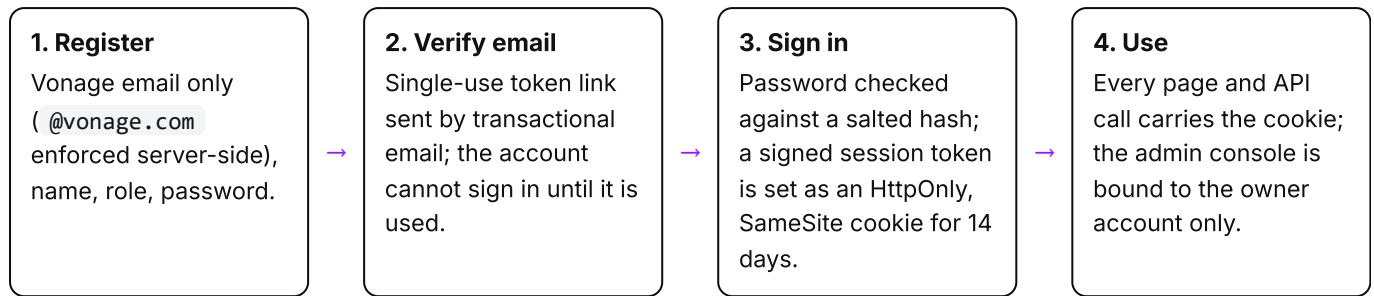
Messaging, Verify, Voice, Video and Number Insight tiles call the real APIs with real numbers and return real message ids. Only the Network APIs and Identity tiles are client-side simulators. The catalogue marks each tile with `mock:true|false` .

04 Data and state

DOCUMENT	CONTENT	LIFETIME
<code>dw_users</code>	Registered employees: name, role, email, salted password hash, verified flag, verification and reset tokens, last login	Until deleted by the admin
<code>logs</code> , <code>conversations</code>	Every outbound and inbound message with channel, status from delivery receipts, per-contact thread (contact identity = phone digits)	Rolling; capped lists
<code>agents</code> , per-demo docs	Conversational agent definitions, demo enquiries, poll tallies, session state for multi-step demos	Per demo; multi-step flows expire (24 h TTL)
Assets prefix	Generated images, rendered videos, PDFs, shop and landing pages	Public-read, versioned by path

Reads go through a small cache so the hot documents are fetched once per process; writes update the cache and S3 together. No personal data beyond the employee's name, role, Vonage email and the phone numbers they choose to demo to.

05 Authentication and access control



- **Password recovery:** forgot-password link, single use, 60-minute expiry; a completed reset also proves the mailbox and verifies the account. Responses never reveal whether an address exists.
- **Resend verification** for accounts stuck before their first sign-in.
- **Admin:** user list, activation and deletion, restricted to the platform owner.
- **Transactional email:** authenticated SMTP with multipart messages and full send logging; a second sender (Gmail API, OAuth) is available as fallback.

06 Webhooks: how the round trip works

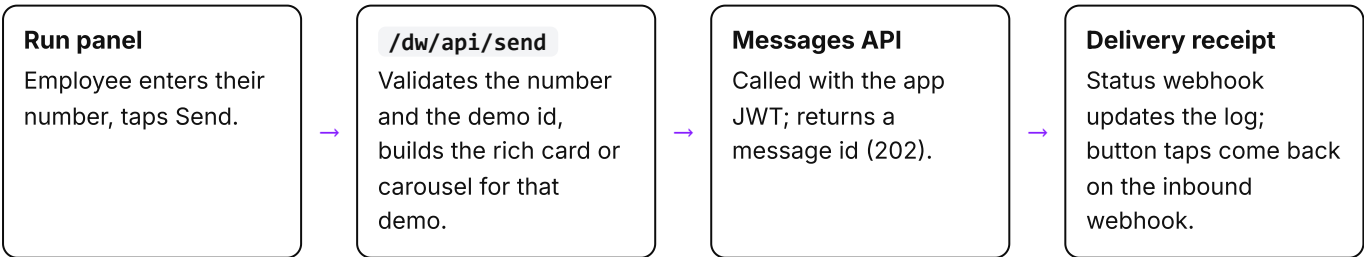
Demo-World both sends and receives. Outbound calls to Vonage are authenticated with an application JWT (RS256, private key held on the server, one-hour expiry) or with the account key and secret where the API requires it. Inbound traffic from Vonage arrives on public webhook URLs behind CloudFront, on the uncached `/2s/*` and `/dw/*` paths.

WEBHOOK	SOURCE	WHAT HAPPENS
POST <code>/2s/webhooks/inbound</code>	Messages API (RCS, WhatsApp, SMS, Viber replies and button taps)	Acknowledged with 200 immediately, then processed asynchronously: de-duplicated by message id, logged to the contact's thread, copied to the CRM bridges, then routed through an ordered chain of handlers (opt-out keywords, demo modules, conversational agents). The first handler that claims the message stops the chain, so demos never fire on each other's triggers.
POST <code>/2s/webhooks/status</code>	Messages API delivery receipts	Updates the stored message with submitted, delivered, read, rejected or expired; rejections are logged with the carrier error for investigation.
GET <code>/dw/voice/answer</code> , <code>/dw/voice/conf</code> , <code>/dw/voice/rtc-answer</code>	Voice API answer URLs	Return NCCO for the call: text-to-speech alerts, IVR menus, conference bridges, SIP connections to AI agents, in-browser calls via the Client SDK.
POST <code>/dw/voice/rtc-event</code> , event URLs	Voice and RTC events	Call state (ringing, answered, completed) is logged for the demo's result panel.
Webview callbacks (<code>/dw/peak/*</code> and similar)	Demo webviews opened from RCS cards	Form submissions and game results post back and the module replies into the same chat thread.

Where Vonage sends the webhooks

Each Vonage application (messaging agent, voice, video, RTC) points its inbound, status, answer and event URLs at this instance. Because several products share applications with other tools, a webhook map page shows the live targets and can re-assert them in one call.

Outbound flow example: send an RCS demo



07 Security protocols

Transport and edge

- HTTPS only through CloudFront with TLS 1.2+; HTTP requests are redirected. HSTS is set on the distribution.
- Static assets cached at the edge; API and webhook paths bypass the cache so no personal response is ever shared between users.
- Security headers on every response: Content-Security-Policy restricting scripts to the platform and its allow-listed SDK hosts, X-Frame-Options, Referrer-Policy and X-Content-Type-Options.

Identity and sessions

- Registration limited to Vonage email addresses and gated by mailbox verification; passwords stored as salted hashes (Argon2id parameters), minimum length enforced.
- Session cookie is signed with a server-side secret, HttpOnly and SameSite so it is unreadable by scripts and not sent on cross-site requests, which also mitigates CSRF for the JSON APIs.
- Login throttling: progressive delay and temporary lockout after repeated failures per account and per source address; all auth events are logged.
- Reset and verification tokens are random 144-bit values, single use, time-boxed.

Vonage and third-party credentials

- All secrets (Vonage API secret, application private keys, AI keys, mail password) are Elastic Beanstalk environment variables or key files packaged at deploy time; nothing is committed to source control. Keys are rotated on a schedule and immediately when a person with access leaves.
- Vonage calls use short-lived RS256 application JWTs; Basic auth is used only where the API requires it and always over TLS.
- Inbound webhooks are verified: signed webhooks (JWT in the Authorization header, HS256 with the account signature secret) are validated before processing, and unsigned or stale payloads are dropped.
- Least-privilege IAM: the runtime role can read and write only the platform's S3 prefixes and invoke Bedrock; deployment credentials are separate from runtime credentials.

Abuse and cost controls

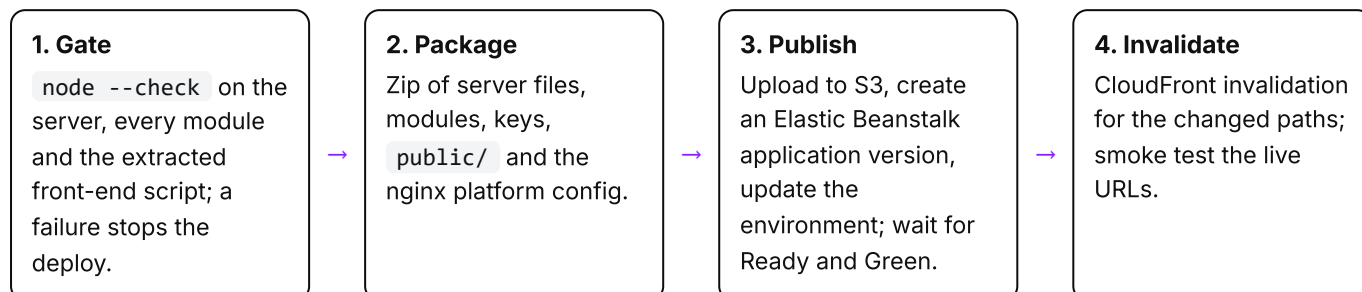
- Every send endpoint validates the demo id against the catalogue and the destination as E.164, and refuses short or malformed numbers.
- Per-user and per-IP rate limits on the send, call and verify endpoints, plus a daily cap per user, so a leaked session cannot become a pumping vector.
- Demos run on a dedicated Vonage sub-account with a spend cap, Fraud Defender country rules and burst protection enabled, and alerts routed to the platform owner.
- WhatsApp's 24-hour rule is respected by design: cold WhatsApp demos are conversational (the user messages first) rather than pushed.
- Audit trail: every outbound demo is logged with the employee, destination, channel, message id and delivery status; the admin can see who sent what.

Data protection

- Data minimisation: no customer PII is stored; demo destinations are the employees' own numbers. Multi-step demo state expires automatically.

- S3 state prefix is private, encrypted at rest (SSE-S3), accessible only to the runtime role; only the asset prefix is public.
- Dependencies are scanned on every deploy and the Node runtime is kept on the current LTS line; the pre-deploy gate also syntax-checks the front-end bundle so a broken page cannot ship.

08 Deployment and operations



- **Observability:** application logs are tailed from Elastic Beanstalk; every demo module logs with its own prefix, every webhook hit and every mail send is logged with its outcome. CloudWatch alarms on 5xx rate and instance health page the owner.
- **Rollback:** any previous application version can be redeployed in one call; state lives in S3 so rollback never loses data.
- **Capacity:** one instance serves the field team comfortably; the design is horizontally scalable because state is external, with the only caveat that video session ids and the webhook de-duplication window would move to a shared store.

09 Demo catalogue by category

CATEGORY	TILES	VONAGE PRODUCTS	MODE
Messaging (17)	RCS skin coach, Wayfair deals, branded RCS builder, Peak Season event (RCS and WhatsApp), ePARK parking, stylist, tracking, abandoned cart, support, carousel, SMS reminder and marketing, three Viber demos	Messages API, Verify, Voice, AI vision	Live
Verify and 2FA (4)	SMS OTP, WhatsApp OTP, voice OTP, risk-based flow	Verify v2, Number Insight, Identity Insights	Live
Voice (7)	IVR, web call, WhatsApp call, alert, survey, AI agent call, conference bridge	Voice API, Client SDK, WhatsApp Business Calling, Bedrock	Live
Video (2)	Telehealth, video banking	Video API	Live
Number Insight (1)	Lookup	Number Insight	Live
Network APIs (7)	SIM swap, silent auth, device swap, location, location verify, reachability and roaming, quality on demand	Identity Insights, Network APIs	Simulator
Identity (3)	Identity Insights, Fraud Defender, subscriber match	Identity Insights, Fraud Defender	Simulator
Integrations (2)	Salesforce, HubSpot	CTI, messaging, voice	Live connectors
Internal tools (7)	Opt-out manager, video maker, Viber capability tour, Android app and others	Various	Tools